



Counted Twice, Caught Once

What a human pill check still catches beyond the robot's own fill log

Kwame Lindqvist · Casimir Beng — School of Emergent Priorities

Background

Dispensing robots log their own weight-based fill accuracy for every prescription. Pharmacies also require a human pre-handover pill count on top. Little is known about how much of that second check's value the first one hasn't already banked.

Aims

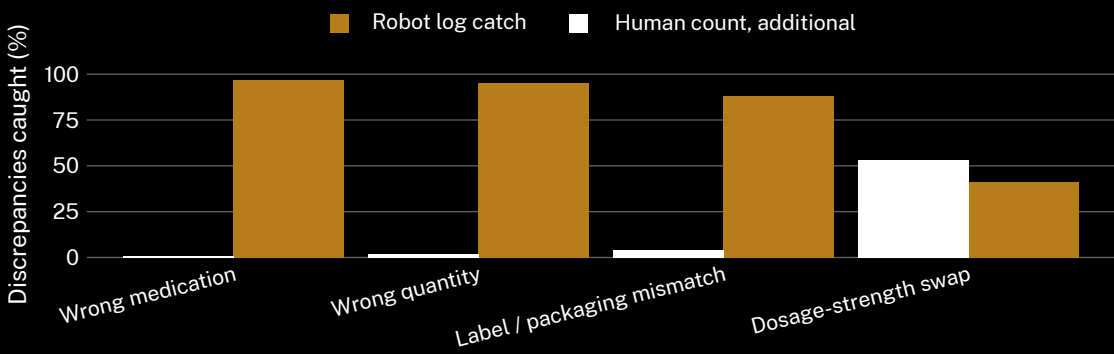
- Quantify how many additional discrepancies the human count catches beyond the robot's own log
- Test whether independent catches concentrate in a specific error type
- Compare that concentration against a chain policy applying the same check at every handover regardless

Methods

- 42 pharmacy sites · merged robot-log and human-count records · 10-month window (Apr 2024–Jan 2025)
- 2,918 flagged discrepancies classified by error type; catch attributed to whichever check logged it first
- Site staffing and count protocol held constant across the window

Results

The human count independently catches an additional 6.1% of flagged discrepancies overall; 71% of those additional catches sit in one category the robot's own log records as a correct fill.



Independent human-count catches by error type, 42 pharmacy sites, April 2024–January 2025 (n = 2,918 flagged discrepancies). Dosage-strength swaps account for the large majority of catches the robot's own log would otherwise have missed.

Discussion & conclusions

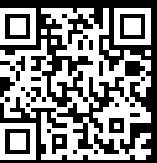
A blanket verification layer applied uniformly returns almost all of its marginal value from one narrow, identifiable failure class; elsewhere it mostly reconfirms what the log already knows. Next: a count directed only at dosage-strength-ambiguous fills, rather than every handover.

References

1. Parasuraman, R., & Manzey, D. H. (2010). Complacency and bias in human use of automation: An attentional integration. *Human Factors*, 52(3), 381–410. doi:10.1177/0018720810376055
2. Lyell, D., & Coiera, E. (2017). Automation bias and verification complexity: A systematic review. *Journal of the American Medical Informatics Association*, 24(2), 423–431. doi:10.1093/jamia/ocw105
3. Skitka, L. J., Mosier, K. L., & Burdick, M. (1999). Does automation bias decision-making? *International Journal of Human-Computer Studies*, 51(5), 991–1006. doi:10.1006/ijhc.1999.0252
4. James, K. L., Barlow, D., McArtney, R., Hiom, S., Roberts, D., & Whittlesea, C. (2010). Incidence, type and causes of dispensing errors: A review of the literature. *International Journal of Pharmacy Practice*, 17(1), 9–30. doi:10.1211/ijpp.17.1.0004
5. Poon, E. G., Cina, J. L., Churchill, W., Patel, N., Featherstone, E., Rothschild, J. M., Keohane, C. A., Whittemore, A. D., Bates, D. W., & Gandhi, T. K. (2006). Medication dispensing errors and potential adverse drug events before and after implementing bar code technology in the pharmacy. *Annals of Internal Medicine*, 145(6), 426–434. doi:10.7326/0003-4819-145-6-200609190-00006

De-identified fill logs only; no prescriber or patient identifiers retained.

Office of Research Outputs · slop.university



“The log says correct. The count disagrees — in one place only.”

Independent catches concentrate in a single dosage class across 42 pharmacy sites.